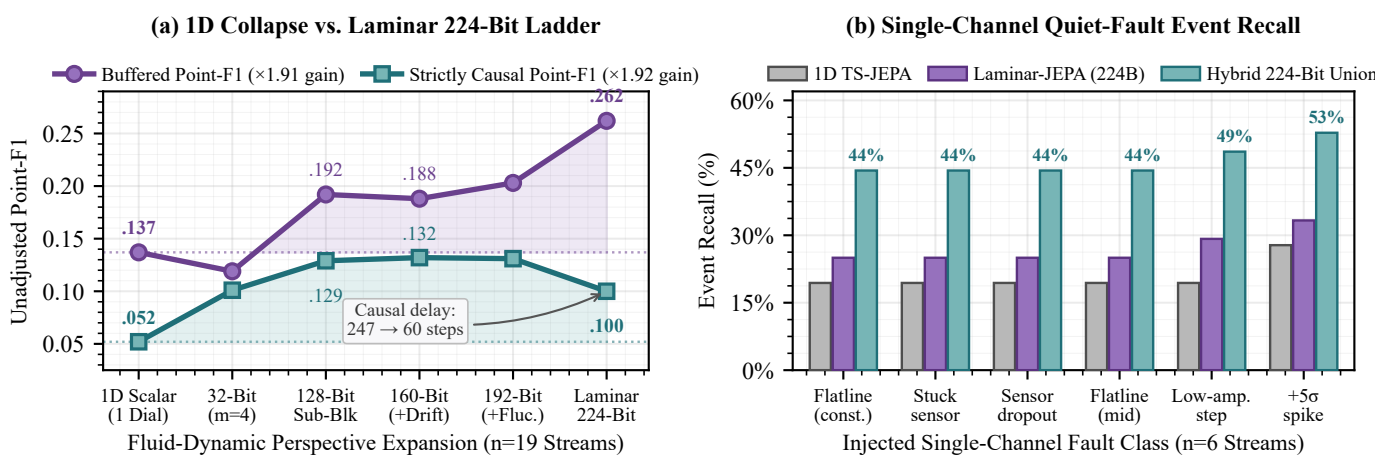

(a) 1D Collapse vs. Laminar 224-Bit Ladder
Buffered Point-F1 (×1.91 gain)
Strictly Causal Point-F1 (×1.92 gain)
Unadjusted Point-F1
0.25
0.20
0.15
0.10
0.05
.137
.052
.192
.188
.132
.129
.262
.100
Causal delay:
247 → 60 steps
1D Scalar
(1 Dial)
32-Bit
(m=4)
128-Bit
Sub-Blk
160-Bit
(+Drift)
192-Bit
(+Fluc.)
Laminar
224-Bit
Fluid-Dynamic Perspective Expansion (n=19 Streams)
(b) Single-Channel Quiet-Fault Event Recall
1D TS-JEPA
Laminar-JEPA (224B)
Hybrid 224-Bit Union
Event Recall (%)
60%
45%
30%
15%
0%
44%
44%
44%
44%
49%
53%
Flatline
(const.)
Stuck
sensor
Sensor
dropout
Flatline
(mid)
Low-amp.
step
+5σ
spike
Injected Single-Channel Fault Class (n=6 Streams)